Dynamic Social Optimum: Braess Network

COMP3821

1 Definition and reduction to path assignments

The simulator uses linear edge costs $c_e(f) = a_e + b_e f$ and the total social cost

$$SC = \sum_t \sum_{e \in E} f_e(t)\, c_e\big(f_e(t)\big),$$

where $f_e(t)$ is the number of agents that traverse e during timestep t. The *dynamic social optimum* is the lowest value of this expression over every feasible way of routing the agents:

$$\boxed{SC_{\text{opt}} = \min_{\text{feasible routing decisions}} \sum_t \sum_{e \in E} f_e(t)\, c_e\big(f_e(t)\big)}$$

Feasible routings. The movement rules of the specification (Section 3.1) and the simulator fix what "feasible" means:

1. all N agents are released at $t = 0$ (none of these scenarios uses cascades);
2. at every timestep each unfinished agent traverses exactly one edge, so no agent can wait;
3. with `forward_only` (the default), an agent never moves to a node with smaller hop distance from its origin and never revisits a node.

Lemma 1 (Routing = path assignment)**.** *Let $\mathcal{P}_{od}$ be the simple paths from o to d that obey rule 3. Every feasible routing gives each agent a path $P_i \in \mathcal{P}_{o_i d_i}$, and agent i traverses the k-th edge of P_i at timestep $t = k - 1$. Every assignment of such paths can be realised. Therefore*

$$SC = \sum_{e \in E} \sum_{t \geq 0} X_{e,t}\,\big(a_e + b_e X_{e,t}\big), \qquad X_{e,t} = \#\{\, i : e \text{ is the } (t+1)\text{-th edge of } P_i \,\},$$

and SC_{opt} is the minimum of this over all path assignments.

Proof. Rule 2 means an agent's position at time t is the end of the t-th edge it has taken. Rule 3 means the edges it takes form a simple path in the allowed subgraph. So $f_e(t) = X_{e,t}$. Conversely, a central planner can tell each agent which next edge to commit to at each step. The outcome depends only on the paths, so every path assignment is achievable. □

So the dynamic problem is a *static* problem on the *time-expanded* resources (e, t). Agents on the same edge only slow each other down if they cross it in the same step.

Lemma 2 (When the dynamic and static optima coincide)**.** *Suppose every edge e sits at the same position $k(e)$ on every feasible path that contains it. Then $X_{e,t} = x_e\,\mathbf{1}[t = k(e) - 1]$, where x_e is the total number of agents using e. So $SC = \sum_e x_e\, c_e(x_e)$, which is the classical static social-optimum objective (Beckmann; Roughgarden) restricted to integer flows.*

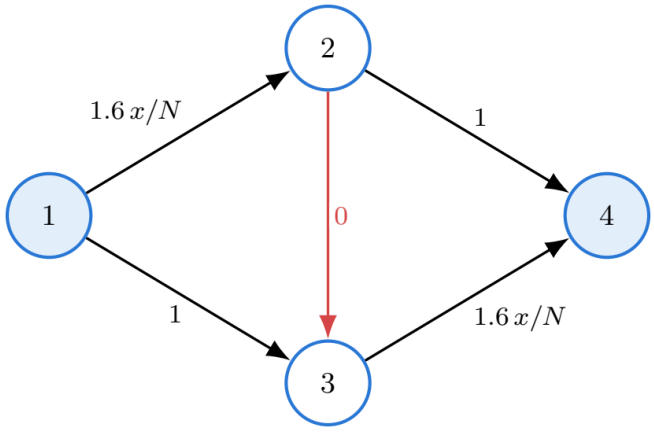
Integer versus continuous. Agents are indivisible, so SC_{opt} is an integer program. Relaxing the path counts to non-negative reals gives a convex quadratic program. Its minimum is a lower bound on SC_{opt} and equals the limit of SC_{opt}/N for large N. Both are reported below. Because c_e is linear, the cost one extra agent adds to resource (e, t) is

$$\Delta^{+}_{e,t} = (X+1)\,c_e(X+1) - X\,c_e(X) = a_e + b_e(2X+1), \qquad X = X_{e,t},$$

and in the continuous relaxation the marginal cost is $\frac{d}{dX}\big[X c_e(X)\big] = a_e + 2b_e X$.

Scaling with N. Every small scenario sets $b_e = \beta_e/N$ with β_e fixed, so the per-agent optimum SC_{opt}/N is almost the same for $N = 20, 50, 80$.

2 Network



Edge	a_e	b_e	$\beta_e = b_e N$
$1 \to 2$	0	$1.6/N$	1.6
$2 \to 4$	1	0	0
$1 \to 3$	1	0	0
$3 \to 4$	0	$1.6/N$	1.6
$2 \to 3$	0	0	0

Scenario	N	$b_{12} = b_{34}$
`braess_N20`	20	0.08
`braess_N50`	50	0.032
`braess`	80	0.02

Write $b = 1.6/N$. All agents travel $1 \to 4$. The three feasible paths are

$$U = 1 \to 2 \to 4, \qquad L = 1 \to 3 \to 4, \qquad Z = 1 \to 2 \to 3 \to 4 \text{ (the zig-zag)},$$

with u, ℓ and z agents respectively, where $u + \ell + z = N$.

3 Derivation

Timing. Lemma 1 gives the resources each path uses:

Path	$t = 0$	$t = 1$	$t = 2$
U	$1 \to 2$	$2 \to 4$	–
L	$1 \to 3$	$3 \to 4$	–
Z	$1 \to 2$	$2 \to 3$	$3 \to 4$

Edge $3 \to 4$ is the *second* edge of L but the *third* edge of Z. Lemma 2 therefore does **not** apply. The ℓ agents on L cross $3 \to 4$ at $t = 1$ and the z agents on Z cross it at $t = 2$, so the two groups never congest each other there.

Objective. The non-zero loads are $X_{12,0} = u + z = N - \ell$, $X_{13,0} = \ell$, $X_{24,1} = u$, $X_{34,1} = \ell$ and $X_{34,2} = z$. Hence

$$\begin{aligned} SC &= (u+z)\, b(u+z) \;+\; \ell \cdot 1 \;+\; u \cdot 1 \;+\; \ell\,(b\ell) \;+\; z\,(bz) \\ &= b(N-\ell)^2 + b\ell^2 + bz^2 + (N - z), \end{aligned}$$

using $u + \ell = N - z$. The only coupling constraint is $u = N - \ell - z \geq 0$.

The objective separates. Apart from $u \geq 0$, SC is a sum of a convex function of ℓ and a convex function of z:

$$g(\ell) = b(N-\ell)^2 + b\ell^2, \qquad h(z) = bz^2 - z.$$

- $g'(\ell) = -2b(N-\ell) + 2b\ell = 0 \Rightarrow \ell^* = N/2$.
- $h'(z) = 2bz - 1 = 0 \Rightarrow z^* = \dfrac{1}{2b} = \dfrac{N}{3.2} = 0.3125\,N$.
- Then $u^* = N - \ell^* - z^* = 0.1875\,N \geq 0$, so the constraint does not bind.

Because the constraint is slack, the unconstrained minimiser is the constrained one. The optimal value is

$$SC^* = 2b\frac{N^2}{4} + \Big(bz^{*2} - z^*\Big) + N = \frac{bN^2}{2} + N - \frac{1}{4b} = 0.8N + N - \frac{N}{6.4} = 1.64375\,N.$$

Integer optimum. Each of g and h is a quadratic, so each is minimised at the integer nearest its vertex. For even N, $\ell = N/2$ exactly and $z = \text{round}(N/3.2)$. The constraint $u \geq 0$ stays slack in all three cases, so these choices are jointly optimal.

4 Result

N	u (U)	ℓ (L)	z (Z)	SC_{opt}	SC_{opt}/N	Continuous bound /N
20	4	10	6	32.880	1.64400	1.64375
50	9	25	16	82.192	1.64384	1.64375
80	15	40	25	131.500	1.64375	1.64375

At $N = 80$, an agent on U pays $0.02 \cdot 40 + 1 = 1.8$, an agent on L pays $1 + 0.02 \cdot 40 = 1.8$, and an agent on Z pays $0.8 + 0 + 0.02 \cdot 25 = 1.3$. Total: $15(1.8) + 40(1.8) + 25(1.3) = 131.5$.

$$\boxed{SC_{\text{opt}}/N \to 1.64375 \quad (\ell = N/2,\ z = N/3.2,\ u = 0.1875N)}$$

5 Comparison with the static (literature) optimum

In the static model, edge $3 \to 4$ carries $\ell + z$ agents whenever they cross it, so

$$SC_{\text{static}} = b(u+z)^2 + b(\ell+z)^2 + u + \ell.$$

By symmetry take $u = \ell = (N - z)/2$. Then $SC_{\text{static}} = \frac{b}{2}(N+z)^2 + N - z$, and

$$\frac{d\,SC_{\text{static}}}{dz} = b(N+z) - 1 = 1.6\Big(1 + \frac{z}{N}\Big) - 1 > 0 \quad \text{for all } z \geq 0.$$

So the static optimum sends nobody on the zig-zag: $u = \ell = N/2$ and $SC_{\text{static}} = 1.8N$ (36, 90 and 144 for $N = 20, 50, 80$).

The step-based simulator can do better, $1.64375N$, because agents on Z reach $3 \to 4$ one step after the L agents and do not add to their congestion. Using the shortcut is *socially* useful in this model, unlike in the classical paradox. Moving from the static routing to the dynamic one shifts $0.3125N$ agents from U/L to Z and saves $0.15625N$.

Verification. The values were checked three independent ways, and all three agree exactly: an exact convex min-cost-flow solver on the time-expanded network, the continuous relaxation, and single-agent local search.